A Baseline Study for Text-Line Detection in Handwritten Traditional Mongolian Archives

Lanying Liang[1] and Yuefeng Liu[2*]

[1]Archives, Inner Mongolia University of Science and Technology, Baotou, 014010, Inner Mongolia, China.
[2*]School of Digital and Intelligent Industry (School of Cyber Science and Technology), Inner Mongolia University of Science and Technology, Baotou, 014010, Inner Mongolia, China.

*Corresponding author(s). E-mail(s): liuyuefeng@imust.edu.cn;
Contributing authors: lianglanying@imust.edu.cn;

Abstract

Text-line detection is a prerequisite for handwriting recognition in historical document pipelines, yet no prior work has addressed this task for handwritten traditional Mongolian archives—a vertically written, cursive, and degraded script. We present a baseline study with a documented dataset of 799 development pages (700 training, 99 validation) and 200 test pages, containing 20,115 text-line polygon annotations converted to bounding-box format. We train YOLOv8n and evaluate with greedy IoU@0.5 matching. On the 200-page test set, YOLOv8n achieves F1 = 0.969 (P = 0.947, R = 0.992), with similar mAP50 on validation and test (0.993 on both). Bootstrap 95% CI for F1 is [0.962, 0.975]. We additionally evaluate YOLOv8n from random initialization (F1 = 0.944) and YOLOv8s (F1 = 0.959), and document the annotation protocol, inter-annotator agreement, and source-stratified data split. Classical projection-based heuristics were attempted but yielded F1 $<$ 0.15. We note that polygon-to-bbox conversion limits stricter metrics (mAP50-95 = 0.607), and that further baselines and polygon-aware evaluation are needed before a complete benchmark can be claimed. The dataset, evaluation scripts, and trained weights are prepared for release subject to institutional image access constraints.

Keywords: text-line detection, historical document analysis, traditional Mongolian script, YOLOv8n, dataset protocol

1 Introduction

Text-line detection—localizing individual lines of text on a document page—is a foundational step for handwritten text recognition and reading-order recovery. For modern printed documents, robust detectors are available. For historical handwritten scripts, particularly low-resource and vertically written ones such as traditional Mongolian, the task has received almost no attention.

Traditional Mongolian is written vertically from top to bottom, in columns flowing left to right. Archival pages contain handwritten text on aging paper with seals, stains, rotated scans, and heterogeneous sources. These conditions make off-the-shelf text-line detectors unreliable.

Our prior work established column-level detection for these archives (F1 = 0.875) [1] and diagnosed the oracle-to-real gap in learned reading-order recovery [2]. Text-line detection is the natural intermediate step: columns must be decomposed into lines before any handwriting recognition model can operate. This paper provides a documented baseline and dataset protocol for that task.

The contributions are: (1) a documented text-line detection dataset of 999 annotated traditional Mongolian archive pages with 20,115 polygon labels, including annotation protocol, inter-annotator agreement data, and source-stratified split documentation; (2) YOLOv8n, YOLOv8s, and YOLOv8n-from-scratch baselines; and (3) a diagnostic analysis of polygon-to-bbox conversion trade-offs and source-stratified evaluation.

2 Related Work

Text-line detection for historical Latin-script documents has been extensively studied [3]. The READ-BAD dataset [4] established detection benchmarks for archival manuscripts, and deep learning methods including ARU-Net [5], dhSegment [8], and fully convolutional approaches have become standard. The ICDAR competition series on complex document layouts (e.g., RDCL2019) [7] provides systematic evaluation protocols for multiple document types. For Asian vertical scripts, text-line detection has been explored for Chinese historical manuscripts [10] and Japanese classical documents [11], but traditional Mongolian—with its distinctive cursive vertical script, manuscript degradation, and lack of existing text-line benchmarks—remains unaddressed at the text-line level.

Column-level layout analysis for Mongolian archives was addressed by [1] using YOLOv8n and Faster R-CNN detectors. The polygon annotations used in this work were originally created for handwriting recognition and are repurposed here for text-line detection. YOLOv8n [9] and YOLOv8s are chosen as baseline detectors for their speed, reproducibility, and strong performance on small custom datasets.

Table 1 Archive sources. The test set uses pages from the same folders as development (source-stratified split, not held-out archive).

Source folder	Train+Val	Test	Total
Folder A (80-48 series)	410	–	410
Folder B (expansion)	240	100	340
Folder C (additional)	149	100	249
Total	799	200	999

3 Dataset

3.1 Source and Composition

The dataset consists of 999 handwritten traditional Mongolian archive page images with polygon-level text-line annotations. Images are high-resolution scans (median $\sim$6,794 $\times$ 4,830 pixels) from multiple archive folders (Table 1). Pages span multiple decades and exhibit diverse degradation types (ink fading, paper yellowing, seal impressions, scanning artifacts).

3.2 Dataset Splits

Pages were split into 700 training, 99 validation, and 200 test pages. We refer to the training and validation pages together as the *development set* (799 pages). The test set draws pages from the same archive folders as the development set but uses disjoint page IDs (source-stratified random split). True held-out archive evaluation with entirely distinct source folders is left to future work.

3.3 Annotation Protocol

Polygon annotations (one per text line, average 12 vertices) were created by two annotators using a standardized guideline. Text lines were defined as continuous horizontal strips of ink across a single column; short fragments (<3 characters) and isolated stains were excluded. A 50-page subset (stratified by archive source and visual degradation level) was annotated independently by both annotators working in isolation. Inter-annotator agreement, measured as polygon-based F1 at 0.75 poly-IoU (vertex-level matching), was 0.841 (precision 0.876, recall 0.812). Disagreements were concentrated on ambiguous short lines ($\sim$45% of disagreements) and boundary cases where ink bleed connects adjacent lines ($\sim$35%). Disagreements were resolved by an adjudication round: annotators reviewed each disagreement case side-by-side and reached consensus; remaining unresolved cases (<2%) were decided by an independent third annotator. All adjudicated cases were then included in the final dataset.

3.4 Polygon-to-Bounding-Box Conversion

Polygon annotations were converted to axis-aligned bounding boxes by computing the bounding rectangle of each polygon vertex set. This conversion is lossy for curved or

Table 2 Dataset statistics after polygon-to-bbox conversion.

Split	Pages	Text lines	Lines/p.	Unique sources
Train	700	∼14,500	20.7	2
Val	99	∼2,026	20.5	2
Test	200	3,615	18.1	2
Total	999	20,115	20.1	3

skewed text lines, which are common in handwritten Mongolian. The resulting dataset contains 20,115 bounding boxes across 999 pages (Table 2). The average page contains 21.1 text lines (range: 5–44).

4 Experimental Setup

4.1 Baselines

YOLOv8n and YOLOv8s. YOLOv8n (3.2M parameters) and YOLOv8s (11.1M parameters), both initialized from public COCO-pretrained weights. Training: 100 epochs, image size 960, batch size 1, SGD momentum 0.937, cosine learning rate schedule (initial 0.01), device: Apple M5 Pro (MPS). Default YOLOv8 augmentations (mosaic, HSV jitter). Single class: `TextLine`. Confidence threshold 0.35 for evaluation.

YOLOv8n from scratch. To quantify the benefit of COCO pretraining, we also train YOLOv8n from random initialization with identical hyperparameters. This provides a lower-bound reference for the pretraining contribution.

Additional baselines (planned). We acknowledge that ARU-Net [5], dhSegment [8], and Faster R-CNN variants would strengthen the baseline portfolio; these are planned for a follow-up comparative study. Classical projection-based heuristics were attempted but yielded F1 < 0.15 due to the vertical script layout, confirming the task is non-trivial for non-learning methods.

4.2 Evaluation Metrics

Primary metric: greedy IoU@0.5 matching (identical protocol to [1]). We report precision, recall, F1, and mean matched IoU. Bootstrap 95% CIs use 1,000 page-level resamples. We also report COCO-style mAP for reference, and discuss the mAP50-95 gap as a diagnostic of the polygon-to-bbox conversion trade-off.

5 Results

Table 4 reports the main results. COCO-pretrained YOLOv8n achieves F1 = 0.969 with high recall (0.992). Training from random initialization reduces F1 to 0.944, confirming that COCO pretraining provides a meaningful initialization benefit (+2.5 pp). YOLOv8s (11.1M parameters vs. 3.2M for YOLOv8n) achieves F1 = 0.959, slightly below the smaller model, suggesting that additional capacity does not improve performance on this dataset size. Similar mAP50 on validation and test (0.993 for YOLOv8n)

Table 3 Validation vs. test comparison for YOLOv8n (COCO pretrained).

Split	P	R	F1	mAP50
Val (99 p.)	0.984	0.980	0.982	0.993
Test (200 p.)	0.947	0.992	0.969	0.993

Table 4 Main results on the 200-page test set.

Method	P	R	F1	mAP50	mAP50-95
Projection heuristic	–	–	*<0.15*	–	–
YOLOv8n (random init)	0.906	0.985	0.944	–	–
YOLOv8n (COCO pretrain)	0.947	0.992	**0.969**	0.993	0.607
YOLOv8s (COCO pretrain)	0.928	0.992	0.959	0.994	0.612

indicates consistent performance across the source-stratified split, though F1 decreases from 0.982 (validation) to 0.969 (test), reflecting higher false positives on the test pages (1.0 FP/page vs. 0.5 FP/page on validation). Bootstrap 95% CI for YOLOv8n test F1 is [0.962, 0.975].

5.1 Polygon-to-Bbox Diagnostic

The gap between mAP50 (0.993) and mAP50-95 (0.607) is substantial. This reflects the polygon-to-bbox conversion: at stricter IoU thresholds, rectangular boxes poorly approximate the curved extent of handwritten Mongolian text lines. The ∼39-point mAP drop mirrors findings in ICDAR competitions where polygon-based evaluation is preferred for curved text [7]. Future work should evaluate polygon-aware metrics directly on the original annotations.

5.2 Error Analysis

The 200 false positives (1.0 per test page) are primarily located at page edges (scanning borders: ∼40%), near seal impressions (∼25%), at ink bleed between lines (∼20%), and in other regions (∼15%). The 30 false negatives are concentrated on very short text lines (≤3 words, ∼60%) and severely faded ink segments (∼40%). Incorporating the Ignore-region protocol from [1] could reduce edge and seal FPs, but would not affect the inter-line bleed FPs, which reflect a fundamental ambiguity in polygon-to-bbox conversion.

5.3 Comparison with Column Detection

In prior work, YOLOv8n column detection achieved F1 = 0.875 on a 43-page training set [1]. The text-line F1 of 0.969 is higher, but this comparison is confounded by unequal training set sizes (700 vs. 43 pages), different test sets, and different target granularities. A controlled comparison with matched training budgets and identical splits is needed before drawing conclusions about relative task difficulty.

6 Discussion

Positioning. This paper is intentionally a *dataset/protocol note and baseline report*, not a comprehensive method comparison or complete benchmark. The contributions are the documented annotation protocol with IAA data, the converted YOLO-format dataset, and the reproducible baseline results. While the F1 = 0.969 result demonstrates the feasibility of text-line detection on these archives, several elements are intentionally left to future work: (1) comprehensive baseline comparison (ARU-Net, dhSegment, Faster R-CNN, and segmentation-based methods remain to be evaluated); (2) polygon-native evaluation to complement bbox-based metrics; (3) true held-out cross-archive evaluation; and (4) larger annotated test set for finer-grained error analysis. A qualitative failure figure (edge FP, seal FP, ink bleed, short-line FN) would also strengthen future iterations.

Annotation limitations. The IAA of 0.841 (polygon-based) indicates moderate annotation consistency. Disagreement sources include ambiguous short-line boundaries and inter-line ink bleed. Future annotation rounds should include more double-annotated pages, explicit boundary guidelines for ambiguous cases, and quality-control adjudication records. The current polygon-to-bbox conversion discards precise shape information and cannot be precisely reversed.

Reproducibility and release. The converted YOLO-format labels, training configuration, evaluation scripts, and trained model weights are prepared for release under institutional data-sharing agreements. The original polygon annotations and archive images are subject to institutional access constraints; reproduction of training requires authorized image access, while metric recomputation from shared prediction files is fully reproducible.

7 Conclusion

We present a baseline study and documented dataset protocol for text-line detection in handwritten traditional Mongolian archives. YOLOv8n achieves F1 = 0.969 (COCO pretrained) and 0.944 (random init), demonstrating the value of pretraining. Classical projection-based heuristics yield F1 < 0.15, confirming the task is non-trivial for non-learning methods. Similar mAP50 on validation and test (0.993) supports reproducibility, and bootstrap CI [0.962, 0.975] bounds the uncertainty. Key remaining work includes: adding segmentation-based and two-stage detection baselines; evaluating polygon-native metrics to quantify the bbox conversion loss; evaluating on truly held-out archive sources; and increasing the annotated test set size. The dataset protocol, evaluation scripts, and trained weights are prepared for release.

Data Availability

Converted YOLO-format labels, training configuration, evaluation scripts, and trained weights are prepared for release. The original polygon annotations and archive images are subject to institutional access constraints; metric recomputation from shared prediction files is fully reproducible without image access.

Author Contributions

Lanying Liang: Data Curation, Annotation, Writing – Original Draft. Yuefeng Liu: Methodology, Software, Validation, Writing – Review & Editing, Supervision.

Ethics Statement

This study uses historical archive images for document analysis and digital preservation. No human-subject experiment is involved. Historical seals and institutional markings are treated as layout artifacts and not used to extract personal identifiers. Example figures have been approved for publication by the hosting archive institution.

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Conflict of Interest

The authors declare no competing interests.

References

[1] L. Liang, Y. Liu, "Pilot-Scale Text Column Detection and Reading-Order Recovery for Traditional Mongolian Historical Archives," submitted, 2026.

[2] L. Liang, Y. Liu, "Reading-Order Recovery Under Detection Noise in Historical Mongolian Document Analysis," submitted, 2026.

[3] L. Likforman-Sulem, A. Zahour, B. Taconet, "Text line segmentation of historical documents: a survey," *Int. J. Document Analysis and Recognition*, vol. 9, pp. 123–138, 2007.

[4] T. Grüning, R. Labahn, M. Diem, F. Kleber, S. Fiel, "READ-BAD: A new dataset and evaluation scheme for baseline detection in archival documents," arXiv:1705.03311, 2017.

[5] T. Grüning, G. Leifert, T. Strauß, J. Michael, R. Labahn, "A two-stage method for text line detection in historical documents," *Int. J. Document Analysis and Recognition*, vol. 22, pp. 285–302, 2019.

[6] K. Nikolaidou, M. Seuret, H. Mokayed, M. Liwicki, "A survey of historical document image datasets," *Int. J. Document Analysis and Recognition*, vol. 25, pp. 305–338, 2022.

[7] C. Clausner, A. Antonacopoulos, S. Pletschacher, "ICDAR2019 competition on recognition of documents with complex layouts," in *Proc. ICDAR*, 2019, pp. 1521–1526.

[8] S. Ares Oliveira, B. Seguin, F. Kaplan, "dhSegment: A generic deep-learning approach for document segmentation," in *Proc. ICFHR*, 2018, pp. 7–12.

[9] G. Jocher, A. Chaurasia, J. Qiu, "Ultralytics YOLOv8," software, version 8.0.0, 2023. Available: https://github.com/ultralytics/ultralytics.

[10] Y. Xu, W. He, F. Yin, C.-L. Liu, "Page segmentation for historical handwritten documents using fully convolutional networks," in *Proc. ICFHR*, 2020, pp. 37–42.

[11] T. Clanuwat, M. Bober-Irizar, A. Kitamoto, A. Lamb, K. Yamamoto, D. Ha, "Deep learning for classical Japanese literature," *NeurIPS Workshop on Machine Learning for Creativity and Design*, 2018.

[12] Y. Pan, D. Fan, H. Wu, D. Teng, "A new dataset for Mongolian online handwritten recognition," *Scientific Reports*, vol. 13, article 26, 2023.

[13] Z. Shen, R. Zhang, M. Dell, et al., "LayoutParser: A unified toolkit for deep learning based document image analysis," in *Proc. ICDAR*, 2021, pp. 131–146.